

Seafloor Classification by Fusing AUV Acoustic and Magnetic Data: Toward Complex Deep-Sea Environments

Zhengren Zhu¹, Chunhui Tao, Tao Wu², Jens Schneider von Deimling, Jinhui Zhang, and Guoyin Zhang

Abstract—Mid-ocean ridges, seamounts, and back-arc basins are focal points for deep-sea research due to their abundant mineral and biological resources. Unlike the sediment-covered deep-sea plains, the above areas have complex topography due to frequent geological activity, which has greatly constrained the accuracy of seafloor classification using shipborne underwater acoustic remote sensing. With the development of autonomous underwater vehicles (AUVs), near range measurements become feasible. Here, we present a method for fusing acoustic and magnetic AUV data on a meter scale, thereby providing evidence for improved seafloor classification. First, a space scale normalized model was built for obtaining high spatial resolution seafloor magnetic anomalies. In the second step, nine features were extracted from the acoustic backscatter mosaic, the bathymetry, and the seafloor magnetic anomaly. Finally, deep neural network (DNN) models were built for training, testing, and classification. To evaluate the classification performance of the model, the method was applied to the survey of the Chinese contract area for polymetallic sulfide exploration on the Southwest Indian Ridge (SWIR) and validated by field data (seafloor video). The integration of acoustic and magnetic data, as opposed to using single acoustic data, improved the overall classification accuracy and kappa coefficient of DNN for basalts, breccias, and sediment by 6.4% and 0.096%, respectively. The experimental results show that the method can fully mine the acoustic and magnetic properties of the seafloor, effectively respond to the challenges of seafloor classification presented by the deep-sea complex environment, and provide a fresh idea for the research of near-bottom high-precision seafloor classification methods.

Index Terms—Autonomous underwater vehicles (AUVs), backscatter intensity, complex deep-sea environment, magnetic anomaly, seafloor classification, underwater acoustic remote sensing.

Received 11 December 2024; revised 11 February 2025; accepted 20 February 2025. Date of publication 25 February 2025; date of current version 10 March 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 42127807 and Grant 42276065, in part by the National Key Research and Development Program of China under Grant 2022YFC2805603 and Grant 2022YFC2803904, in part by the Stable Supporting Fund of National Key Laboratory of Underwater Acoustic Technology under Grant JCKYS2023604SSJS016, and in part by China Scholarship Council. (Corresponding authors: Chunhui Tao; Tao Wu.) Zhengren Zhu is with the Ocean College, Zhejiang University, Zhoushan 316021, China.

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Digital Object Identifier 10.1109/TGRS.2025.3545128

I. INTRODUCTION

A. Acoustic Seafloor Classification

SEAFLOOR classification and mapping constitute a critical component of marine environmental surveys [1]. Digitized seafloor classification maps can provide the necessary environmental information for marine spatial planning and management, marine engineering construction, marine scientific research, and beyond [2], [3], [4], [5].

Currently, shipborne underwater acoustic remote sensing has become the preferred technology for characterizing the deep seafloor (see Fig. 1). Acoustic devices, such as multibeam echosounder systems (MBESs) and sidescan sonar systems (SSSs), can be deployed aboard survey vessels to obtain bathymetric and backscatter data, from which multivariate features can be extracted [6]. After that, combining the field data, machine learning algorithms (e.g., support vector machines [7], [8], random forests [9], [10], [11], convolutional neural networks [12], deep neural networks (DNNs) [13], [14], etc.) can map seafloor properties including sediment grain size, seawater permeability, and biological habitats, which have been extensively validated [9], [15].

B. Problem Statement

The deep sea, encompassing approximately two-thirds of the Earth's surface, exhibits a diverse array of seafloor topography and geomorphological features [16]. Among them, the abyssal plains, which constitute roughly 75% of the deep-sea floor, are typically blanketed by extensive deposits of fine-grained sediment, with thicknesses ranging from a few meters to several kilometers (see Fig. 1) [17]. Despite the fact that detection accuracy and resolution of shipborne sonar decrease with increasing water depth, the seafloor classification maps produced for the abyssal plains with flat topography and simple seafloor types are sufficient for various purposes. Unlike the abyssal plains, mid-ocean ridges, seamounts, and back-arc basins exhibit complex topography and seafloor distributions due to frequent geological activities such as volcanism, earthquakes, and submarine landslides (see Fig. 1). Furthermore, the characteristics of deep-sea systems, thousands of meters water deep, the accuracy, and resolution of seafloor classification by traditional shipborne acoustic remote sensing are greatly discounted in these areas. The mid-ocean ridge, seamounts, and back-arc basins, despite being only a small portion of the deep-sea floor, contain most of the deep-sea

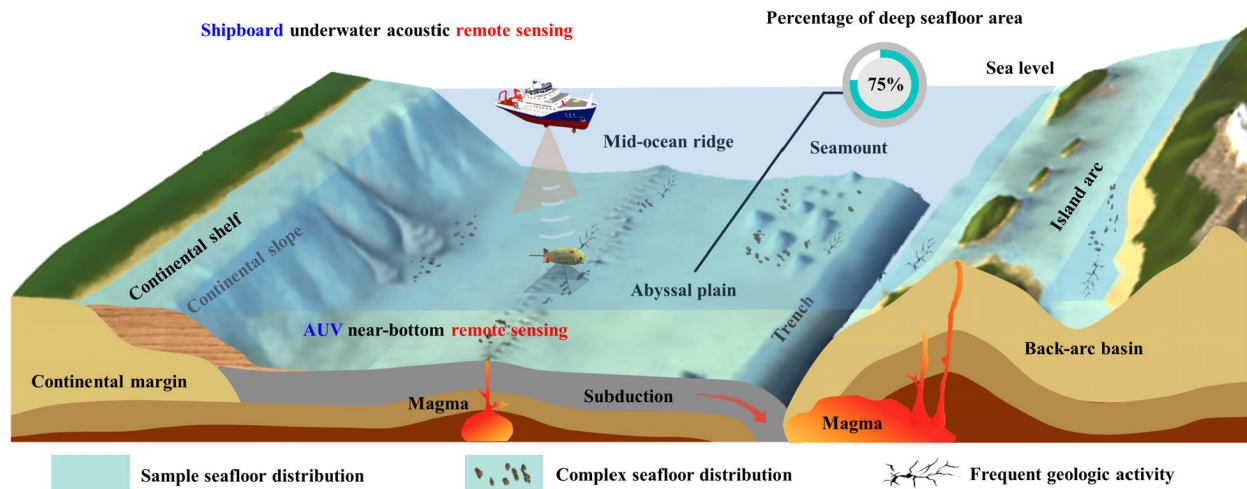


Fig. 1. Interpretation of seafloor differences in different geomorphologies of the deep-sea and seafloor survey methods applicable in different areas.

mineral resources (such as polymetallic sulfides, cobalt-rich ferromanganese crusts, and polymetallic nodules) and deep-sea ecosystems (such as seamount ecosystems and chemosynthetic communities), and are the focus of deep-sea surveys and research [18], [19], [20].

C. Study Idea

Currently, near-bottom detection using acoustic equipment on autonomous underwater vehicles (AUVs) to obtain high-spatial-resolution acoustic backscatter mosaic is one of the most direct and effective technical solutions for dealing with complex deep-sea environments (see Fig. 1) [21], [22]. Considering economic costs and efficiency, AUVs usually integrate multiple devices such as acoustic, magnetic, and chemical sensors for synoptic observations [23]. Magnetometers can measure the secondary abnormal static magnetic field (magnetic anomaly) excited by the Earth's magnetic field [24], [25]. The magnetic susceptibility at depths extending several hundred meters beneath the seafloor can be estimated from observed magnetic data [26]. Association study indicates that magnetic susceptibility may serve as a substitute for grain size index for seafloor classification [27]. For instance, the magnetic susceptibility of marine sediment, typically ranging from $10^1 \times 10^{-6}$ to $10^4 \times 10^{-6}$, is significantly lower compared to that of basalts, which ranges from $10^3 \times 10^{-6}$ to $10^5 \times 10^{-6}$ [27], [28].

The sonar equipment on AUVs typically operates at high frequencies (high angle resolution) to achieve more refined acoustic imaging [29]. High frequencies mean low acoustic penetration (usually a few centimeters in hard rock, depending on the specific working frequency and the acoustic impedance of seafloors). The detection depth of magnetic data can reach hundreds of meters. According to Tobler's first law of geography, within a confined spatial domain, assuming the same formation structure, magnetic anomalies can effectively reflect the heterogeneity of the superficial seafloor [30], effectively compensating for the deficiency of acoustic data in detecting depth.

D. Study Protocol and Objectives

To this end, a seafloor classification strategy fusing AUV acoustic and magnetic data is innovatively proposed for surveying the complex deep-sea environment. This strategy comprises two primary components: spatial scale normalization and fusion classification. Initially, a spatial scale normalization model based on regression analysis was developed to obtain high-spatial resolution seafloor magnetic anomalies. Subsequently, feature extraction was performed on acoustic backscatter mosaic, seafloor DEM, and seafloor magnetic anomalies. Ultimately, a DNN model was constructed for training and classification. The major objectives of the present study were as follows: 1) to fully mine the acoustic and magnetic properties of the seafloor in order to improve classification accuracy in complex deep-sea environments; 2) to provide a framework for seafloor classification based on AUV multisource data fusion, which can serve as a reference for future seafloor classification combined with other AUV survey data; and 3) to evaluate the contribution of acoustic and magnetic features in the seafloor classification process, thereby providing guidance for future seafloor classification work.

II. SURVEY AREA AND DATA

A. Survey Area

The study area is located on the Southwest Indian Ridge (SWIR) at coordinates $50^\circ 27' E$, $37^\circ 39' S$ [see Fig. 2(a)], with a length of 10 km and a width of 6 km, covering a total area of about 60 km² [see Fig. 2(d)]. Geophysical studies reveal that large-scale magmatism occurred in the study area between 8 and 10 Ma, with the younger oceanic crust being shallower (approximately 1700 m deep) and thicker (9.5 km) than the older oceanic crust [31], [32]. Frequent geologic activity created a rugged topography with numerous individual volcanic cones, fault cliffs, and fissures in the study area. Volcanic cones are defined as cone-shaped volcanic edifices ranging from 500 m to several kilometers in diameter and up to several hundred meters in height [33]. The rock types in

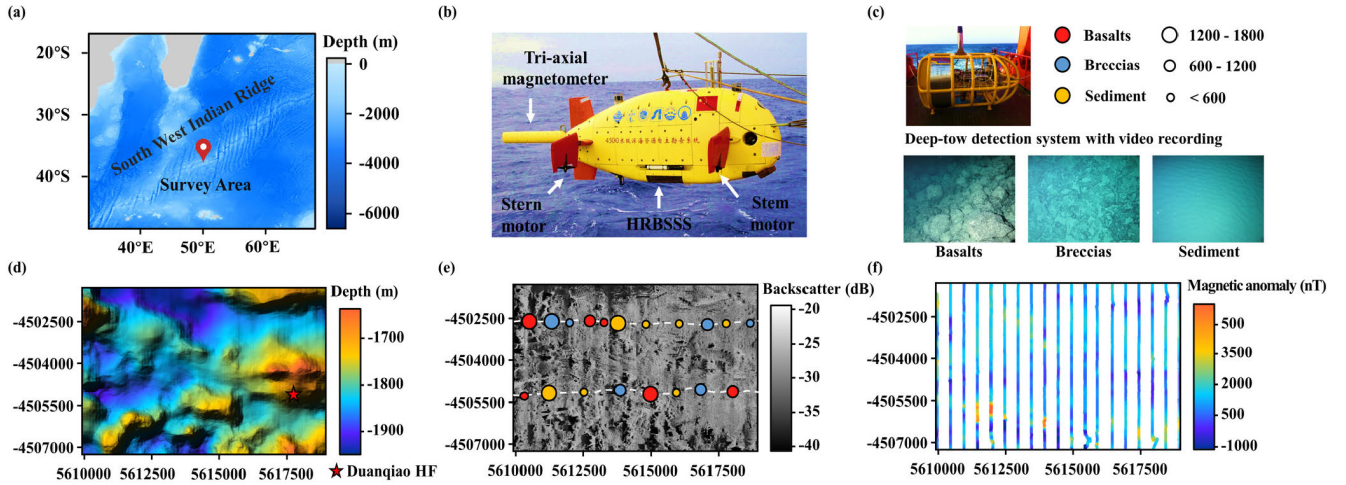


Fig. 2. Location of the study area and experimental data. (a) Location of the study area in the Southwestern Indian Ocean Ridge. (b) Tri-axial magnetometer and the HRBSSS installed on the QianLong-II AUV. The positions of the motor, tri-axial magnetometer, and HRBSSS are marked. (c) Deep-tow detection system and seafloor images of basalts, breccias, and sediment. (d) Seafloor DEM acquired by sonar system equipped on AUV. The red star is the location of the Duanqiao hydrothermal field. The planimetry coordinate datum is WGS_1984_UTM_Zone_39S. (e) Backscatter mosaic acquired by the sonar system equipped on AUV and ground-truth sampling data. The white line shows the track of the deep-tow detection system. Red, blue, and yellow circles represent sample locations of basalts, breccias, and sediment, respectively, and the size of the circle represents the number of samples. (f) Magnetic anomaly acquired by magnetometer equipped on AUV.

the area are predominantly basalts, including pillow, lobate, sheets, and crush-formed breccias [34]. Moreover, the Duanqiao hydrothermal field is located on the western side of the study area and is encircled by a substantial concentration of polymetallic sulfides [35].

B. Experimental Data

1) *AUV Acoustic and Magnetic Data*: In February 2016, during the DY40 cruise, R/V Xiangyanghong-10 carried the QianLong-II AUV that conducted a survey at 50.4°E on the SWIR. AUV is equipped with a high-resolution bathymetric SSS (HRBSSS) and a tri-axial magnetometer [36]. The AUV was surveyed at a speed of 1–2 kt, maintaining a height of approximately 100 m above the seafloor, and following a track spacing of 400 m [see Fig. 2(f)]. Navigation is achieved by a combination of inertial navigation system (iXBlue PHINS-6000) and long baseline system [36].

The HRBSSS is independently developed by the Institute of Acoustics of the Chinese Academy of Sciences, and can simultaneously collect water depth and backscatter intensity data [37]. The water depth measurement accuracy meets the IHO S-44 I standard [37]. The HRBSSS operates at a frequency of 150 kHz and has a swath width of 250 m. Sound speed data are provided by a CTD carried on the AUV. The CleanSweep software was employed to process the water depth and backscatter data, resulting in the generation of a backscatter mosaic [see Fig. 2(e)] and a seafloor DEM [see Fig. 2(d)], both with a spatial resolution of 2 m. The processing of bathymetric data consists of four parts: sound speed correction, outlier rejection, georeferencing, and mosaicking. The backscatter data processing consists of five parts: georeferencing, geometric correction, radiometric corrections (including the effects of gain and water-column propagation), angle dependence removal, and mosaicking.

The tri-axial magnetometer was jointly developed by the Second Institute of Oceanography, Ministry of Natural Resources of China, and Hangzhou Dianzi University. The magnetometer has the characteristics of small size, lightweight, and magnetic probe separation, and is primarily employed for high-precision magnetic surveys near the bottom on AUVs and remotely operated vehicles (ROVs). The sampling frequency of the magnetometer is 1 Hz. Magnetic data processing includes AUV magnetic effect correction and calibration. The magnetic effect of the AUV is estimated and corrected using the magnetic data collected during the AUV rotation. A magnetometer is moored in a fixed location to serve as a temporary geomagnetic station, correcting the International Geomagnetic Reference Field (IGRF) by eliminating the diurnal variation from the measured magnetic field [38]. Finally, the resultant magnetic anomaly data are smoothed using a 7-s moving average filter (about 10 m averaging) [see Fig. 2(f)].

2) *Seafloor Video*: The seafloor video was acquired using the deep-towed detection system aboard R/V Dayangyihao during the DY34 cruise in April 2015 [39]. Ultrashort baseline positioning is used in navigation. The trajectory of the deep-tow system is depicted in Fig. 2(e) as the white dotted line. The seafloor types in the study area are divided based on grain size, from large to small, into basalts, breccias, and sediment [see Fig. 2(c)].

Ideally, the orientation of the camera lens within the deep-towing system should be perpendicular to the seafloor. The actual coverage (number of grids) of the seafloor photograph can be calculated by using the height h of the camera from the seafloor and the camera field of view θ (see Fig. 3). Twenty areas with the same seafloor type were selected as sampling stations [see Fig. 2(e)].

$$\text{Num}_i = \left\{ \frac{\sqrt{2} \tan \theta \bar{h}_i \cdot L_i}{2^2}, i = 1, 2, \dots, 20 \right\} \quad (1)$$

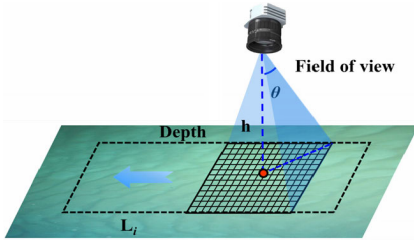


Fig. 3. Statistics method for the sample numbers of seafloor stations.

where Num_i represents the number of samples at the i th station and h_i represents the average height of the camera from the seafloor at the i th station (sampled at 1-s intervals). $\sqrt{2} \tan \theta h_i$ is the average width of the i th station, and L_i represents the distance (length) that the camera moves at the i th station.

III. METHODOLOGY

Borrowing from terrestrial remote sensing, multisource data fusion needs to solve problems such as discrepancies in spatial resolution and coverage, as well as the geographic position correction [40]. Considering that acoustic and magnetic devices are carried on the AUV for simultaneous investigation (with the same detection coverage and coordinate system), the unification of spatial resolution emerges as a critical factor for the fusion of acoustic and magnetic data.

A. Spatial Scale Normalization

A pulse from the acoustic transducer can obtain hundreds of water depth values in a plane orthogonal to the trajectory. Therefore, acoustic remote sensing data are usually high spatial resolution grids covering the entire survey area (see Figs. 2(e) and 4). For different design purposes, the magnetometer employs a “point-line” measurement method for surveys, resulting in magnetic data usually being point clouds with high resolution along the track direction (e.g., 1.3 m herein) (see Figs. 2(f) and 4).

When magnetic data are used for classification, interpolation is required to generate a grid with high spatial resolution (e.g., 2 m herein). Due to the insufficient point cloud density of the magnetic data in the direction perpendicular to the trajectory (e.g., 400 m herein), the interpolated grid may exhibit “grid artifacts” (see Section V for a specific comparison), which will inevitably affect subsequent classification. To this end, a spatial scale normalization model based on regression analysis is proposed.

Suppose $\text{BS}_{\text{grd}}^{2\text{m}}(O)$ and $\text{D}_{\text{grd}}^{2\text{m}}(O)$ represent the backscatter mosaic with a spatial resolution of 2 m and the seafloor DEM with a spatial resolution of 2 m, respectively, and $\text{M}_{\text{pcd}}^{1.3\text{m}}(o)$ represents the magnetic anomaly point clouds with a spacing of about 1.3 m along the trajectory direction. $\text{M}_{\text{grd}}^{2\text{m}}(O)$ represents the magnetic anomaly grid with a spatial resolution of 2 m after normalization.

$$\text{BS}_{\text{grd}}^{2\text{m}}(O) = \left\{ \text{BS}_{\text{grd}}^{2\text{m}}(O_{ij}), i = 1, 2, \dots, m; j = 1, 2, \dots, n \right\} \quad (2)$$

$$\text{D}_{\text{grd}}^{2\text{m}}(O) = \left\{ \text{D}_{\text{grd}}^{2\text{m}}(O_{ij}), i = 1, 2, \dots, m; j = 1, 2, \dots, n \right\} \quad (3)$$

$$\text{M}_{\text{pcd}}^{1.3\text{m}}(o) = \left\{ \text{Z}_{12\text{kHz}}^{2\text{m}}(o_k), k = 1, 2, \dots, l \right\} \quad (4)$$

$$\text{M}_{\text{grd}}^{2\text{m}}(O) = \left\{ \text{M}_{\text{grd}}^{2\text{m}}(O_{ij}), i = 1, 2, \dots, m; j = 1, 2, \dots, n \right\} \quad (5)$$

where i and j represent the row and column of the grid data, respectively, and k represents the point number in the point cloud data. m and n are the number of rows and columns of the grid, respectively, and l is the number of points in the point cloud. $O_{ij} = (U_{ij}, V_{ij})$ and $o_k = (u_k, v_k)$. (U_{ij}, V_{ij}) and (u_k, v_k) are the geographic coordinates (longitude and latitude) of the center of the grid and the point cloud, respectively.

The overall flow of the space scale normalization is shown in Fig. 5 and is divided into two stages. First, the spatial trend of $\text{M}_{\text{grd}}^{2\text{m}}(O)$ is predicted through the application of regression analysis on acoustic and magnetic data. Second, the residuals of the regression analysis are interpolated to reduce the error of the trend fit. Therefore, the general form of the spatial scale normalization model is as follows:

$$\text{M}_{\text{grd}}^{2\text{m}}(O) = \text{T}_{\text{grd}}^{2\text{m}}(O) + \text{R}_{\text{grd}}^{2\text{m}}(O) \quad (6)$$

where $\text{T}_{\text{grd}}^{2\text{m}}(O)$ is the trend component and $\text{R}_{\text{grd}}^{2\text{m}}(O)$ is the residual component.

1) *Spatial Trend Simulation*: Backscatter intensity and magnetic anomalies can characterize the acoustic and magnetic properties of the seafloor for classification. The influence of bottom currents and topography is also considered an important factor shaping the seafloor [41]. Therefore, from a data analytical standpoint, a correlation exists between backscatter intensity, depth, and magnetic anomalies. For instance, depressions are more likely to accumulate sediment, meaning that the backscatter intensity and magnetic anomalies in these areas are relatively low. A combination of $\text{BS}_{\text{grd}}^{2\text{m}}(O)$ and $\text{D}_{\text{grd}}^{2\text{m}}(O)$ was found to significantly increase the prediction feasibility of $\text{T}_{\text{grd}}^{2\text{m}}(O)$. Therefore, $\text{BS}_{\text{grd}}^{2\text{m}}(O)$ and $\text{D}_{\text{grd}}^{2\text{m}}(O)$ are used as auxiliary data to predict $\text{T}_{\text{grd}}^{2\text{m}}(O)$. The specific steps are as follows.

Step 1: Data matching to obtain l 3-D arrays.

$$\begin{aligned} V_k &= \left[\text{BS}_{\text{grd}}^{2\text{m}}(U_k, V_k), \text{D}_{\text{grd}}^{2\text{m}}(U_k, V_k), \text{M}_{\text{pcd}}^{1.3\text{m}}(u_k, v_k) \right] \\ (k &= 1, 2, \dots, l) \\ (U_k, V_k) &= \text{find} \left(\min \left[(U_{ij} - u_k)^2 + (V_{ij} - v_k)^2 \right] \right) \\ (i &= 1, 2, \dots, m; j = 1, 2, \dots, n; k = 1, 2, \dots, l) \end{aligned} \quad (7)$$

where V_k is the general form of the array, containing the backscatter intensity $\text{BS}_{\text{grd}}^{2\text{m}}(U_k, V_k)$, the depth $\text{D}_{\text{grd}}^{2\text{m}}(U_k, V_k)$, and the magnetic anomaly $\text{M}_{\text{pcd}}^{1.3\text{m}}(u_k, v_k)$. (U_k, V_k) represents the grid center coordinates that match point $\text{M}_{\text{pcd}}^{1.3\text{m}}(u_k, v_k)$. (U_k, V_k) can be solved by traversing the grid to find the minimum Euclidean distance [see Fig. 4 and (8)].

Step 2: Regression analysis.

Consider $\text{BS}_{\text{grd}}^{2\text{m}}(U_k, V_k)$ and $\text{D}_{\text{grd}}^{2\text{m}}(U_k, V_k)$ in V_k as independent variables and $\text{M}_{\text{pcd}}^{1.3\text{m}}(u_k, v_k)$ as the dependent variable to construct the regression function f .

$$\text{T}_{\text{pcd}}^{1.3\text{m}}(u_k, v_k) = f \left(\text{BS}_{\text{grd}}^{2\text{m}}(U_k, V_k), \text{D}_{\text{grd}}^{2\text{m}}(U_k, V_k) \right) \quad (k = 1, 2, \dots, l) \quad (9)$$

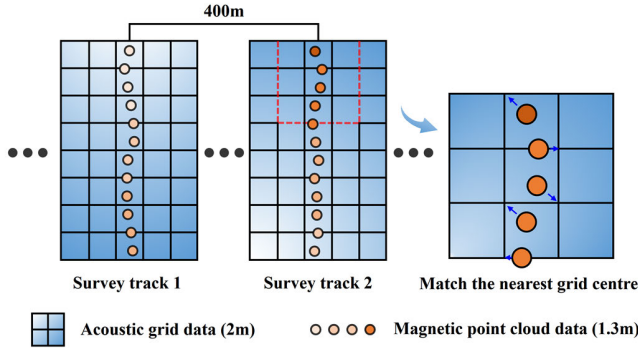


Fig. 4. Schematic of the differences in spatial resolution between acoustic and magnetic data and the method of matching the two.

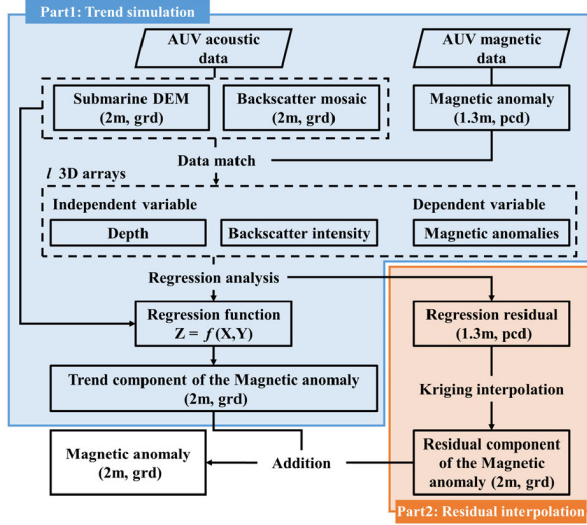


Fig. 5. Flowchart of space scale normalized model.

where $T_{pcd}^{1.3m}(u_k, v_k)$ is the spatial trend of the magnetic anomaly point cloud $M_{pcd}^{1.3m}(o)$.

Step 3: Simulation of spatial trends.

Bring $D_{grd}^{2m}(O_{ij})$ and $D_{grd}^{2m}(O_{ij})$ into f to obtain the spatial trend of $M_{grd}^{2m}(O_{ij})$.

$$T_{grd}^{2m}(O_{ij}) = f\left(BS_{grd}^{2m}(O_{ij}), D_{grd}^{2m}(O_{ij})\right) \quad (i = 1, 2, \dots, m; j = 1, 2, \dots, n). \quad (10)$$

2) *Regression Residual Interpolation*: If the regression function f is perfect, the obtained $T_{grd}^{2m}(O)$ will be error-free, and the residual $R_{grd}^{2m}(O)$ will be 0. However, in practical applications, residuals are invariably present in regression analysis. Considering that the residual has a spatial structure, the residual $R_{pcd}^{1.3m}(o_k)$ of the magnetic anomaly point cloud in Section III-A1 is meshed to obtain $R_{grd}^{2m}(O)$.

$$R_{pcd}^{1.3m}(o_k) = M_{pcd}^{1.3m}(o_k) - T_{pcd}^{1.3m}(o_k) \quad (k = 1, 2, \dots, l) \quad (11)$$

$$R_{grd}^{2m}(O_{ij}) = \sum_{v=1}^{mn} \left[\lambda_v * R_{pcd}^{1.3m}(o_k) \right] \quad (i = 1, 2, \dots, m; j = 1, 2, \dots, n; k = 1, 2, \dots, l) \quad (12)$$

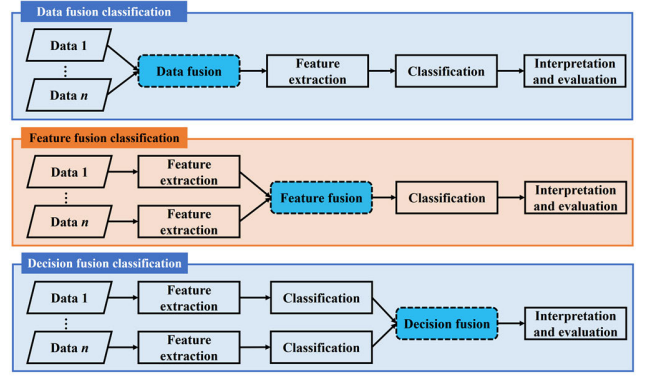


Fig. 6. Flow of three fusion classifications in remote sensing.

where $T_{pcd}^{1.3m}(o_k)$ is calculated from (9). The kriging method is used for interpolation, where λ_v is the interpolation weight coefficient [42]. Finally, according to (6), $T_{grd}^{2m}(O)$ and $R_{grd}^{2m}(O)$ are added to obtain $M_{grd}^{2m}(O)$.

B. Fusion Classification

According to the different stages, the fusion classification in remote sensing is generally divided into three types: data fusion classification, feature fusion classification, and decision fusion classification (see Fig. 6) [43]. Considering that acoustic and magnetic data are obtained from different sensors, a seafloor classification strategy based on feature fusion is employed. This section consists of three parts: feature extraction, classification model construction, and accuracy evaluation.

1) *Feature Extraction*: The seafloor classification features based on grid units can be categorized into two distinct groups: a) mosaic features derived from the backscatter data (e.g., backscatter intensity and gray-level co-occurrence matrix (GLCM) texture features) and b) topography features obtained from bathymetry data, including slope, aspect, mean curvature, roughness, and bathymetry position index (BPI). As summarized in Table I, a total of 17 features were extracted in this experiment, comprising three primary features and 14 secondary features.

The inherent multiscale characteristics of seafloor features present challenges for the comprehensive exploration of seafloor information when employing single-scale feature extraction methods. To address this, multiscale feature extraction was performed on the secondary features listed in Table I, employing various kernel sizes classified as small (3×3 or 1×5), medium (5×5 or 5×10), and large (11×11 or 10×50). It should be noted that the kernel sizes may exhibit slight variations depending on the feature computation method.

The Hughes phenomenon demonstrates that, given a constant number of training samples, the prediction accuracy diminishes as the feature dimensionality increases [48]. Consequently, feature selection is also required before classification. Boruta is presently regarded as one of the most advanced methods for all-relevant feature selection. Boruta evaluates the feature importance by comparing the original features with their permuted counterparts, and is applicable to almost all datasets [49], [50]. This study employs the Boruta algorithm for feature selection, with detailed principles outlined in [50].

TABLE I
SUMMARY OF THE EXTRACTED FEATURES

NO.	The object-extracted	Feature name	Description	Kernel sizes	Primary / Secondary feature
1	Magnetic anomaly grid	Magnetic anomaly	Seafloor magnetic properties	-	Primary feature
2		Backscatter intensity	Seafloor acoustic backscatter intensity	-	Primary feature
3		GLCM-Homogeneity	Homogeneity of GLCM obtained from the backscatter mosaic, representing the uniformity of the image [41].	3x3, 5x5, 11x11	Secondary feature
4		GLCM-Correlation	Correlation of GLCM obtained from the backscatter mosaic, representing the local similarity of the image [41].	3x3, 5x5, 11x11	Secondary feature
5	Backscatter mosaic	GLCM-Contrast	Contrast of GLCM obtained from the backscatter mosaic, representing the sharpness and texture depth of the image [44].	3x3, 5x5, 11x11	Secondary feature
6		GLCM-ASM	Angular second moment (ASM) of GLCM obtained from the backscatter mosaic, representing the texture coarseness of the image [41].	3x3, 5x5, 11x11	Secondary feature
7-9		Hue-Saturation-Intensity	Hue, saturation, and intensity of the backscatter mosaic. These HSI values are transformed into red, green, and blue (RGB) color space, resulting in three variables [45].	3x3, 5x5, 11x11	Secondary feature
10		Moran's I	Spatial auto-correlation in a neighborhood [46]	3x3, 5x5, 11x11	Secondary feature
11		Depth	Elevation value of the surface unit.	-	Primary feature
12		Slope	Steepness of the surface unit [41].	3x3, 5x5, 11x11	Secondary feature
13		Aspect	Direction of slope change [41].	3x3, 5x5, 11x11	Secondary feature
14	Seafloor DEM	Roughness	Inhomogeneity of the surface unit [9].	3x3, 5x5, 11x11	Secondary feature
15		Mean curvature	Degree of bending of the surface unit [47].	3x3, 5x5, 11x11	Secondary feature
16		BPI	A measure of where a georeferenced location, with a defined depth, is relative to the neighboring seascape [41].	1x5, 5x10, 10x50	Secondary feature
17		Complexity	Localized variability in the surface unit [41].	3x3, 5x5, 11x11	Secondary feature

TABLE II
CALCULATION OF ACCURACY EVALUATION INDEX

Classes	Predictions			User's accuracy	Overall accuracy
	Basalts	Breccias	Sediment		
Basalts	A	B	C	$\frac{A}{A+B+C}$	$\frac{A+E+I}{N}$
Breccias	D	E	F	$\frac{D}{D+E+F}$	
Sediment	G	H	I	$\frac{G}{G+H+I}$	
Producer's accuracy	$\frac{A}{A+D+G}$	$\frac{B}{B+E+H}$	$\frac{C}{C+F+I}$	-	
Kappa =					
$\frac{N(A+E+I) - (A+B+C)(A+D+G) - (D+E+F)(B+E+H) - (G+H+I)(C+F+I)}{N^2 - (A+B+C)(A+D+G) - (D+E+F)(B+E+H) - (G+H+I)(C+F+I)}$					

2) *Classification Model Construction*: DNNs represent a canonical architecture in deep learning frameworks [51], [52]. In comparison with shallow neural networks, the DNN multilayer perceptron network structure has better generalization ability and can effectively deal with the challenge of seafloor classification brought by large sample data (such as the 10 640

samples in this study). Therefore, this study employs a DNN model for seafloor classification, with foundational principles detailed in [53].

The DNN architecture (number of hidden layers and number of neurons per layer) is a crucial parameter influencing the quality of the classification. At present, the prevalent approaches for determining the optimal architecture of DNNs involve manual “trial and error” and the evaluation of hyperparameters. In this study, the random search method is employed to assess the hyperparameters with the aim of identifying the optimal DNN model architecture. The rationale of the random search method can be found in the literature [54].

3) *Accuracy Evaluation*: Upon completion of the classification, an accuracy evaluation is conducted by statistically comparing the ground truth labels with the classification results. As the most commonly used evaluation indices in classification tasks, overall accuracy (OA), user's accuracy (UA), producer's accuracy (PA), and kappa coefficient are calculated by confusion matrix [see Table II] [55].

calculated by confusion matrix [55].

Where A, E, and I represent the number of basalt, breccia, and sediment, respectively, that are correctly classified. B and

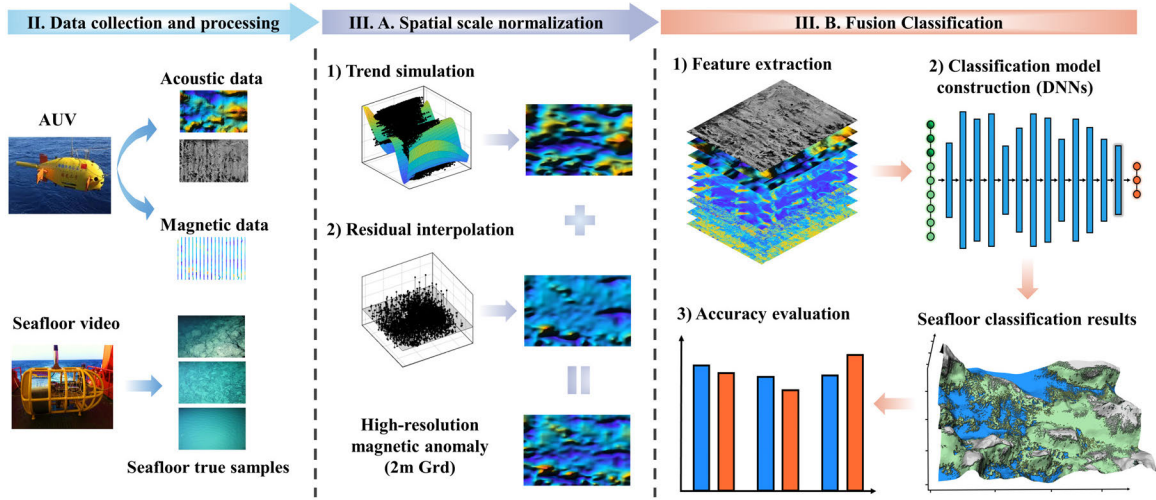


Fig. 7. Workflow of the proposed method.

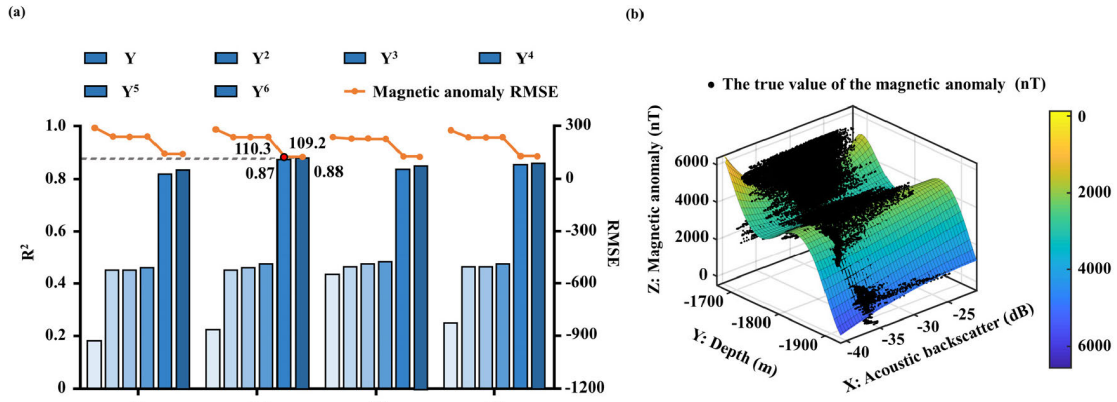


Fig. 8. Regression analysis for 3-D arrays. (a) Error comparison of different polynomial regression analyses. (b) Fitting result of optimal regression analysis.

C represent the number of basalt misclassified as breccia and sediment, respectively. D and F represent the number of breccia misclassified as basalt and sediment, respectively. G and H represent the number of sediments misclassified as basalt and breccia, respectively. $N = A + B + C + D + E + F + G + H + I$ is the total number of test samples.

C. Workflow of the Proposed Method

The overall workflow of the methodology employed in this study is illustrated in Fig. 7. The part of the spatial scale normalization was implemented in MATLAB 2014a. The feature extraction was implemented in ArcMap 10.2. The classification was implemented in Python 3.7 with the “keras” library, and the seafloor mapping was done in Surfer 18 and Microsoft Visio 2019.

IV. RESULTS AND ANALYSIS

A. Spatial Scale Normalization

According to (8), the data matching of the backscatter mosaic, seafloor DEM, and magnetic anomaly point cloud resulted in a total of 87 694 3-D arrays. Randomly select 85 000 arrays and conduct a regression analysis using

backscatter intensity and depth as the independent variables, with magnetic anomaly serving as the dependent variable. The remaining 2694 arrays are used for the algorithm evaluation, as discussed in Section V.

Polynomial regression possesses a flexible nonlinear fitting capability, enabling infinite approximation of the dependent variable by increasing the high-degree terms of the independent variable [56]. Furthermore, polynomial regression has better interpretability compared to black-box models, such as neural networks and random forests [see Fig. 8(b)]. The experiment compared 24 polynomial regression models with different combinations of coefficients and used the coefficient of determination (R^2) and the root mean square error (RMSE) as the evaluation indexes [see Fig. 8(a)]. The smaller the RMSE, the closer the R^2 is to 1, the more accurate the model fit.

As shown in Fig. 8(a), the function exhibits an inflection point at the fifth power of Y (a substantial improvement in the fitting accuracy) and subsequently stabilizes. In order to accommodate the efficiency of the operation, the order of the function is reduced as much as possible with little difference in accuracy. The fitting result ($R^2 = 0.87$ and $RMSE = 110.3$) of the second power of X (backscattering intensity) and the fifth

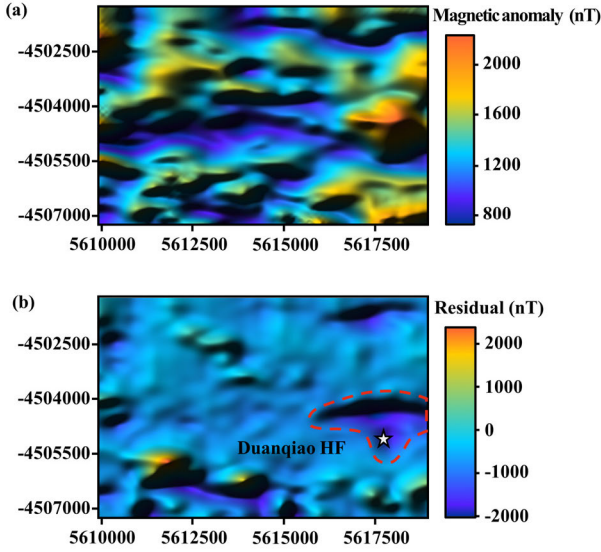


Fig. 9. Trend component $T_{\text{grd}}^2(O)$ and residual component $R_{\text{grd}}^2(O)$ of the magnetic anomaly grid $M_{\text{grd}}^2(O)$. (a) Result of the spatial trend simulation $T_{\text{grd}}^2(O)$. (b) Results of the residual difference $R_{\text{grd}}^2(O)$.

power of Y (depth) is selected as the regression function f .

Magnetic anomaly:

$$\begin{aligned} Z &= f \quad (\text{backscatter: } X \text{ and depth: } Y) \\ &= 1283 - 7.59X + 78.62Y + 3.86X^2 + 7.75XY \\ &\quad + 680.3Y^2 + 44.78X^2Y - 48.93XY^2 - 189.8Y^3 \\ &\quad + 57.67X^2Y^2 - 150XY^3 - 676.8Y^4 + 18.63X^2Y^3 \\ &\quad - 34.47XY^4 - 195.5Y^5. \end{aligned} \quad (13)$$

Bring the backscatter mosaic and seafloor DEM into the function f to obtain the spatial trend component $T_{\text{grd}}^2(O)$ of $M_{\text{grd}}^2(O)$ [see Fig. 9(a)]. The residue of the regression function f is interpolated using kriging at a spatial resolution of 2 m, obtaining the residue component $R_{\text{grd}}^2(O)$ of $M_{\text{grd}}^2(O)$ [see Fig. 9(b)].

Ideally, the residuals from the regression analysis should follow a normal distribution with a mean of 0 in the study area. However, in Fig. 9(b), the area marked by the red dashed line has a large amount of negative residuals, with values spanning approximately from -500 to -2000 nT and an average of -1287 nT [see Fig. 10(a)]. This phenomenon can be attributed to the presence of the Duanqiao hydrothermal field in the area [57]. The high temperatures associated with hydrothermal activity cause the seafloor bedrock to lose its magnetism, resulting in a low magnetic anomaly in the area. Removing the marked area, the overall mean and variance of the residuals are -23 and 163 nT, respectively, returning to normal levels [see Fig. 10(b)].

$T_{\text{grd}}^2(O)$ is added to the $R_{\text{grd}}^2(O)$ via (6) to obtain $M_{\text{grd}}^2(O)$ (see Fig. 11).

B. Fusion Classification

Feature importance analysis was conducted on all secondary features listed in Table I using the Boruta algorithm. The significance of each feature was quantified using the median Z-score (higher values indicate greater importance).

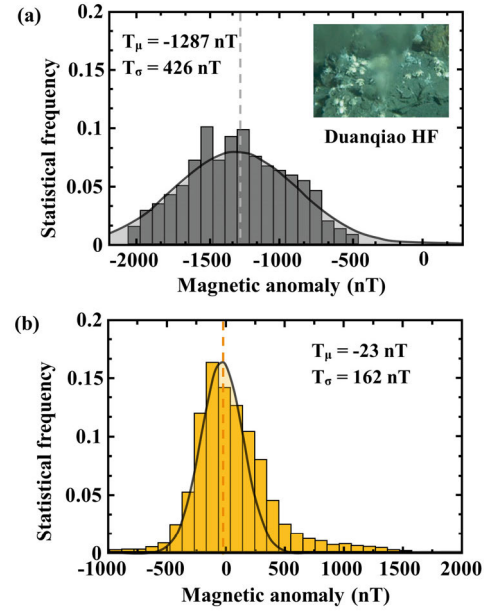


Fig. 10. Results of residual statistics. (a) Residual distribution for the areas marked by the red dashed line in Fig. 9(b). (b) Residual distribution outside the labeled areas in Fig. 9(b).

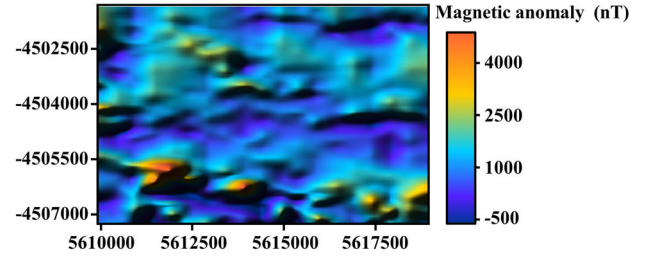


Fig. 11. Magnetic anomaly grid with 2-m spatial resolution.

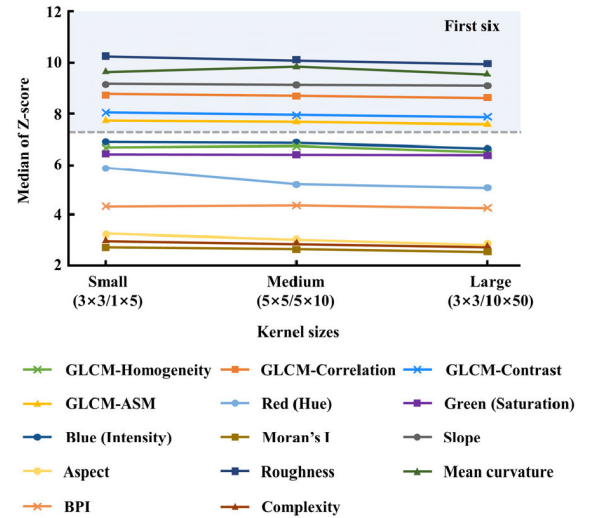


Fig. 12. Importance of secondary features at different scales.

The analysis revealed insignificant variance in feature importance across different scales (see Fig. 12). Except for the mean curvature feature, for which the optimal kernel is of medium size (5×5), the optimal kernels for the other secondary features are small-sized (3×3 or 1×5). These small kernels can effectively constrain the receptive field and preserve the high-frequency details of the seafloor features.

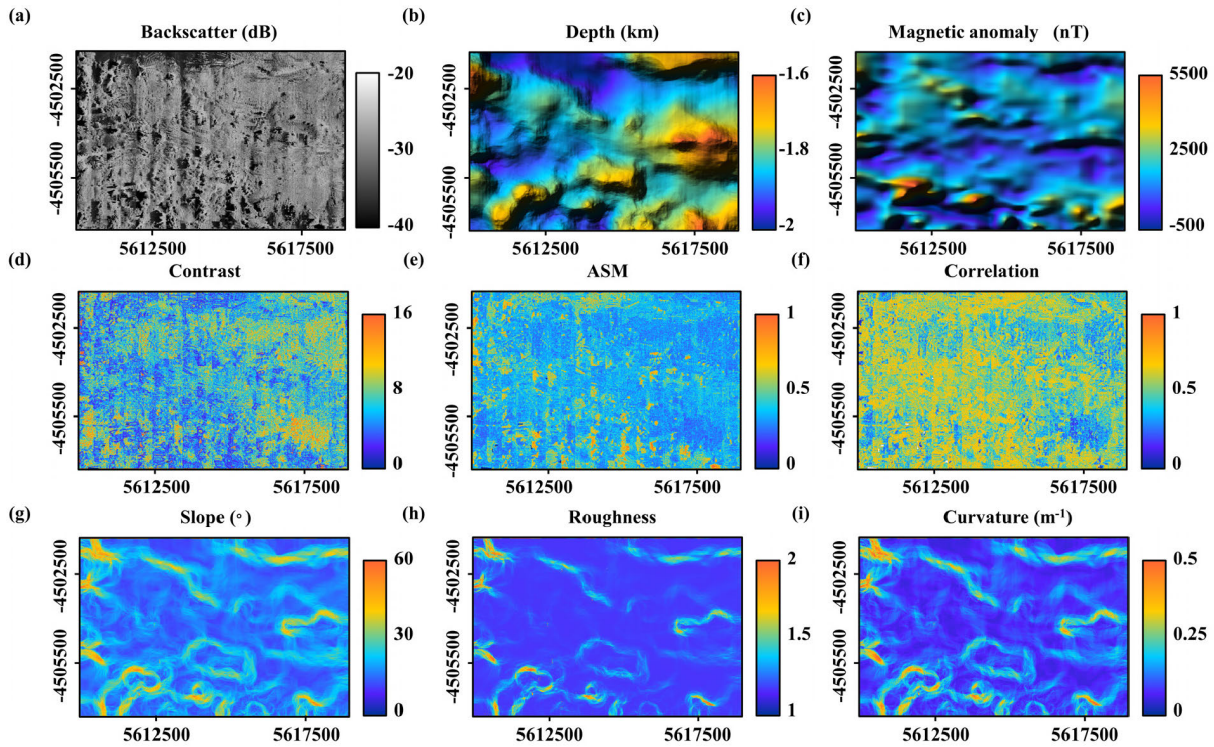


Fig. 13. Features extracted in the study area. (a) Backscatter intensity. (b) Depth. (c) Magnetic anomaly. (d) GLCM-contrast. (e) GLCM-ASM. (f) GLCM-correlation. (g) Slope. (h) Roughness. (i) Curvature.

Ultimately, the six most significant secondary features (highlighted with blue backgrounds in Fig. 12) were selected. Combined with three primary features, a total of nine features were ultimately utilized for seafloor classification (see Fig. 13).

To verify the effectiveness of fusing acoustic and magnetic data for classification, two DNN models with different feature combinations were constructed. The input layer of Model 1 comprises eight features, including four derived from backscatter mosaics and four derived from seafloor topography. The input layer of Model 2 comprises nine features, including four derived from backscatter mosaics, four derived from seafloor topography, and one magnetic anomaly. Table III presents the hyperparameter configuration of the DNNs. To avoid overfitting, a dropout layer was incorporated into the network architecture.

The ground truth sample data are derived from 20 seafloor stations [see Fig. 2(e)], comprising a total of 10 640 grids: 3900 for basalts, 3260 for gravel, and 3480 for sediment. In accordance with the 7:3 principle, 7000 datasets were randomly selected to constitute the training set, while the remaining 3640 datasets were allocated to the test set.

Each model in the hyperparameter configuration was trained for 50 epochs using a random search algorithm. A fivefold cross-validation strategy was adopted for model training and validation (see Fig. 14). The DNN architectures for Model 1 and Model 2 with the highest accuracy were found from an exhaustive search space comprising 80 000 parameter combinations (see Fig. 15). The training set was re-entered into the optimal architectures, and the training was performed for 100 epochs to obtain the final classification models (see Fig. 14).

TABLE III
HYPERPARAMETRIC CONFIGURATION

Hyperparameters	Minimum value	Maximum values	Step
Number of neurons	2	100	2
Number of hidden layers	2	20	2
Dropout	0	0.5	0.05
Learning rate	0.1, 0.01, 0.001, 0.0001		
Activation functions	Rectified Linear Unit (ReLU), Exponential Linear Unit (ELU), Sigmoid, Softmax		
Optimizer	Stochastic Gradient Descent (SGD)		
Loss function	Mean Squared Error (MSE)		
Batch size	20		
Initial bias	Uniform		
Initialization mode	Uniform		

Computational efficiency is an important indicator for evaluating model property. The experimental results show (see Fig. 16) a trade-off between the computational time of the model (including the duration required for parameter optimization via the random search algorithm) and the OA. Model 2 demonstrated an OA of 93.6%, higher than the 86.1% accuracy of Model 1. However, the run time of Model 2 was 3 h and 33 min, longer than the 3 h and 18 min of Model 1. Considering that the time difference between the two models is merely 15 min, Model 2 will have the advantage of higher accuracy if sufficient computational resources are available.

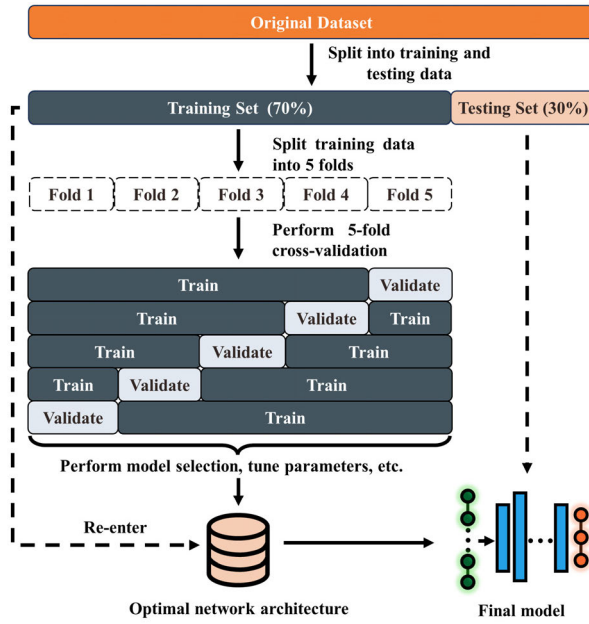


Fig. 14. Model training workflow.

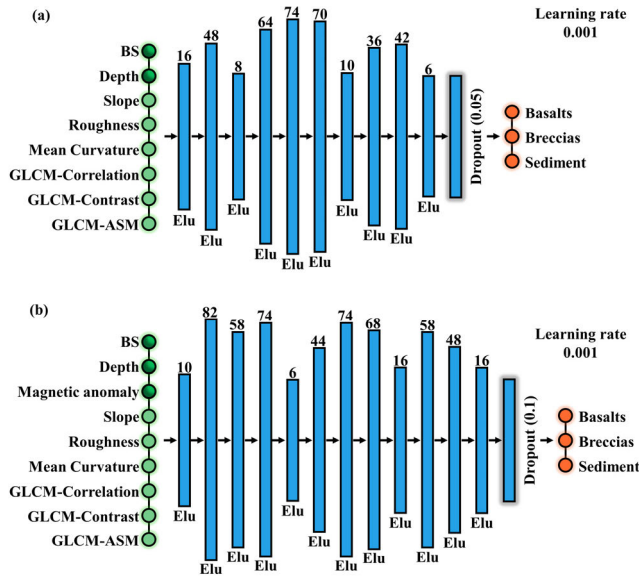


Fig. 15. Two optimal DNN architectures found by the random search algorithm. (a) DNN architecture for Model 1. (b) DNN architecture for Model 2.

Learning and loss curves can also be used to diagnose the learning performance of the models over time (see Fig. 17). The training accuracy curve provides insight into how the model is learning, while the validation accuracy gives insight into the generalizing process. Similarly, the training loss curve shows the model's performance on training data when the optimization algorithm adjusts the model parameters to minimize the loss function. The validation loss curve assesses the model's performance on new unseen data.

Overall, the accuracy and loss curves for Models 1 and 2 show a similar pattern characterized by a sharp increase in accuracy and a sharp decrease in the loss function. This indicates that both models learn quickly and fit the validation

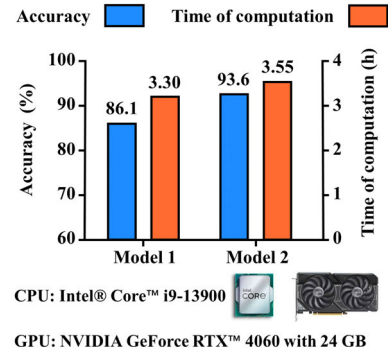


Fig. 16. OA and computational time of the models.

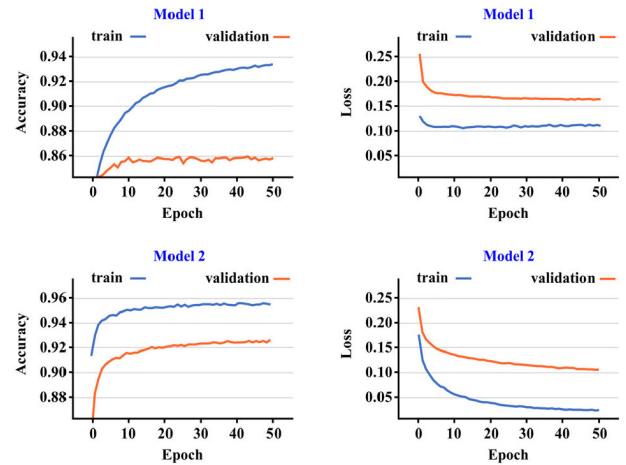


Fig. 17. Learning curve and loss function plots for the models.

data well. Furthermore, the accuracy and loss values of both models stabilize after a certain point, suggesting that their convergence is well resolved.

The classification results indicate that basalts and breccias are the most widely distributed in the study area, mainly located on the higher terrains of mountain peaks and slopes, ranging from -1600 to -1700 m. The sediment is mainly concentrated in the low-lying areas, ranging from -1800 to -1900 m, and is interspersed with some discontinuous broken gravel. The classification results are generally consistent with the typical seafloor distribution patterns observed at mid-ocean ridges [58], [59].

Upon comparing the results of the two classification models, the seafloor classified by fusing magnetic data [see Fig. 18(b)] exhibits a more natural transition than the seafloor classified solely using acoustic data [see Fig. 18(a)]. The classification result of Model 2 obeys the transition sequence from larger to smaller seafloor grain sizes (basalts to breccias to sediment). To quantitatively evaluate the classification accuracy of the models, the test set data were sequentially input into different DNN models for evaluation (see Fig. 19).

Compared to Model 1, Model 2 demonstrates enhanced PA and UA for basalts, breccias, and sediment to varying extents after fusing the magnetic anomaly feature. The OA and kappa coefficient also increased from 84.6% and 0.768 (Model 1) to 91.0% and 0.864 (Model 2), verifying the

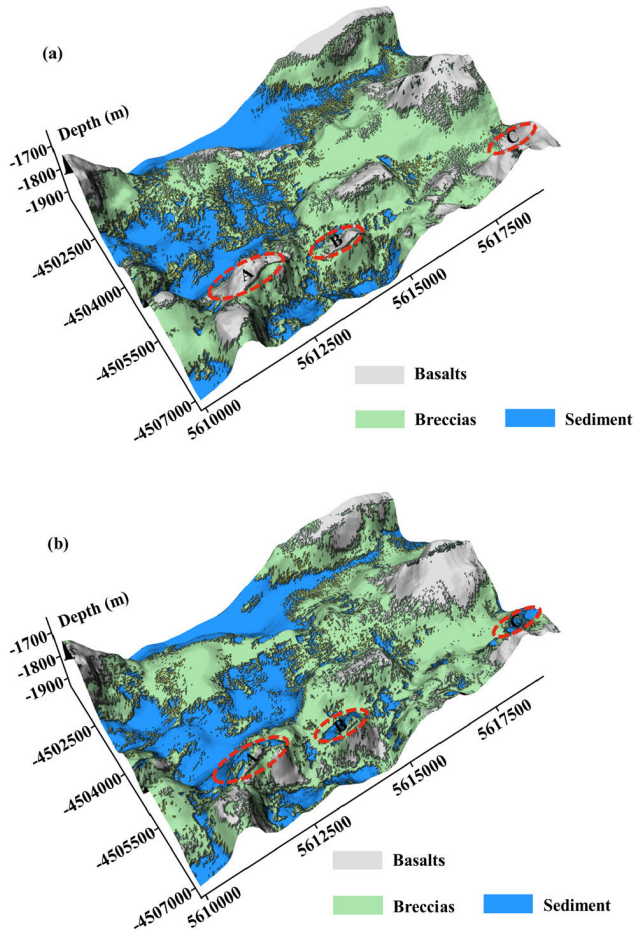


Fig. 18. Seafloor classification maps of different DNN classification models. (a) Seafloor classification map for Model 1. (b) Seafloor classification map for Model 2.

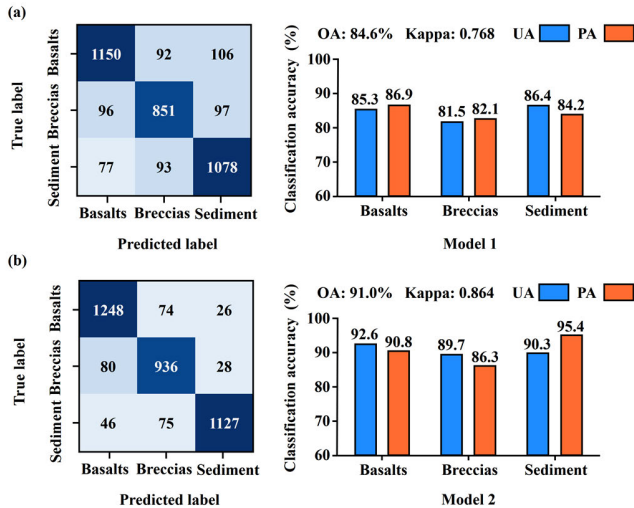


Fig. 19. Accuracy comparison of different DNN classification models. (a) Confusion matrix and accuracy evaluation index for Model 1. (b) Confusion matrix and accuracy evaluation index for Model 2.

effectiveness of fusing acoustic and magnetic data for seafloor classification.

Among all the accuracy evaluation indexes, the PA of sediment exhibited the most significant improvement, increasing from 84.2% in Model 1 to 95.4% in Model 2. The enhancement indicates a substantial reduction in the misclassification

of basalts and breccias as sediment. For a more exhaustive comparison, three sample areas were selected, as marked by the red dotted lines in Fig. 18.

As shown in Fig. 20(a), all three sample areas exhibit the characteristics of high topography. Previous classification results indicate that high topography areas are typically distributed with high acoustic impedance rock substrates, such as basalts and breccias. High impedance means high backscattering intensity. However, the sample area exhibited a low backscattering intensity [see Fig. 20(a)]. The seafloor video in proximity to the sample areas was re-examined. The seafloor fissures and thick sediment layers may account for the low backscatter intensity observed in the sample areas.

In near-seafloor surveys, the acoustic devices deployed by AUVs typically operate at high frequencies, such as 120 kHz in this study, which results in lower acoustic penetration. When the basalts and breccias in the sample areas are covered or semi-covered by thick sediment, the echo intensity received by the sonar system will be significantly diminished [see Fig. 20(b)]. Furthermore, the recurrent seismic activity of the mid-ocean ridge has generated numerous seafloor fissures in the study area [see Fig. 20(b)]. When sound waves reach these fissured seafloor areas, a portion of the sound energy is also lost, reducing echo intensity.

For the classifier, low backscatter intensity means that the seafloor prediction for the sample areas prefers the sediment label. When two diametrically opposed classification-oriented features and their derived features (high terrain and low backscatter intensity) were input into the network of Model 1, the classification uncertainty markedly increased: some basalts and breccias were misclassified as sediment. As mentioned in the introduction, magnetic data possess the capability to detect deeper subsurface and effectively identify buried or semi-buried seafloor rock structures (the sample area in Fig. 20(a) exhibits a high magnetic anomaly). Therefore, for Model 2, which integrates the magnetic anomaly feature, the misclassification of basalts and breccias as sediment is reduced, improving the classification accuracy of all seafloor types.

V. DISCUSSION

A. Evaluation of Space Scale Normalized Model

To facilitate interpretation for various purposes, magnetic data are often gridded using the common difference methods [60]. Here, three interpolation methods—nearest neighbor interpolation (NNI), minimum curvature interpolation (MCI), and kriging interpolation—were employed to generate a magnetic anomaly grid with a spatial resolution of 2 m, in order to evaluate the efficacy of the proposed spatial scale normalization model. The magnetic anomaly data used for interpolation are the same as in Section IV, with a total of 85 000 points. The remaining 2694 points are used to calculate the RMSE for comparative analysis.

Owing to the inherent limitations of the NNI method, which performs sampling without accounting for neighboring grid points, a distinct “mosaic” and “jagged” effect is evident in Fig. 21(a). Even though both the MCI method and the

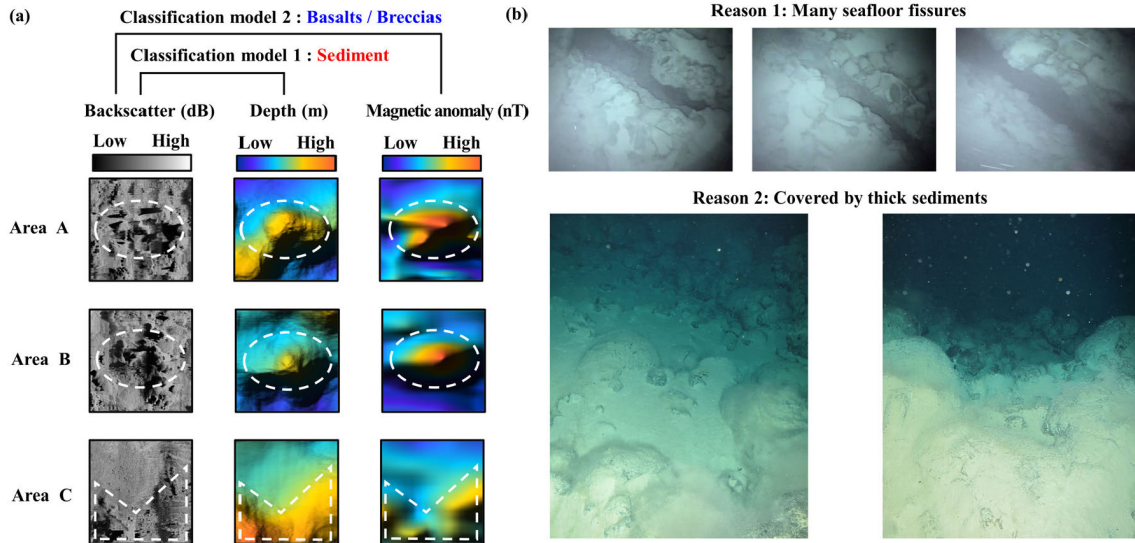


Fig. 20. Comparison of sample areas with differences in classification results and possible reasons for the differences. (a) Comparison of the three primary features of the sample areas. (b) Two possible reasons for differences in classification.

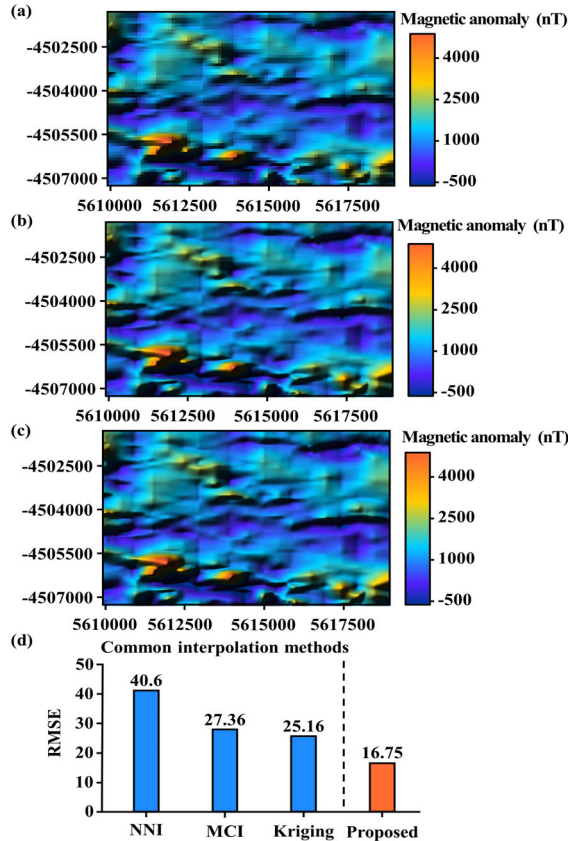


Fig. 21. Comparison of the proposed spatial scale normalization with common interpolation methods in magnetism. (a) Magnetic anomaly grid acquired by NNI. (b) Magnetic anomaly grid acquired by MCI. (c) Magnetic anomaly grid acquired by kriging interpolation. (d) Statistics of RMSE for each method.

kriging interpolation method consider the regional correlation of the data, the highly uneven distribution of the point cloud density (1.3 m in the along-track direction and 400 m in the perpendicular direction) leads to numerous “grid artifacts” and

“strip artifacts” in the interpolation results [see Fig. 21(b) and (c)]. These artificial errors do not compromise the interpretation of magnetic anomalies and related analyses; however, they will undoubtedly affect the classification accuracy when the interpolation results are used for the grid-based seafloor classification.

Comparing the results of the spatial scale normalization (see Fig. 11), the overall transition is smooth and exhibits good continuity. Fig. 21(d) summarizes the RMSE for the four methods. The spatial scale normalization has the highest accuracy, with an RMSE of 16.75, outperforming the NNI (40.6), the MCI (27.36), and the kriging interpolation (25.16), verifying the effectiveness of the spatial scale normalization model proposed in this study.

B. Features Contribution Analysis

As shown in Section IV, fusing magnetic data effectively improves the accuracy of the DNN model for seafloor classification. Therefore, accurately quantifying the contribution of each feature, especially the magnetic anomaly feature, to the classification model is of reference value for guiding future seafloor classification work. Shapley Additive exPlanations (SHAP) is a collaborative game theory-based method to quantify input feature contributions to model predictions, and is one of the most widely used methods for evaluating feature importance in neural networks [61]. Import the SHAP library in Python and use the DeepExplainer class to calculate the SHAP values of each feature. The contribution of each feature to the model is quantified by the average absolute value of the SHAP values for the feature (see Fig. 22).

In the DNN model of this experiment, the contribution of the depth feature is the highest, possibly related to the complex seafloor topography of the mid-ocean ridge. The important influence of hydrodynamics on the distribution and evolution of the seafloor in the study area is highlighted. The contribution of the magnetic anomaly feature is 0.87,

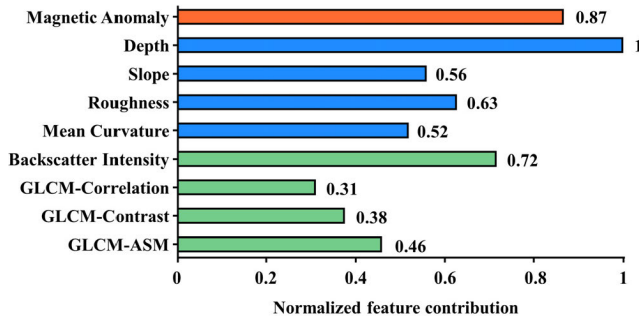


Fig. 22. Contribution of features to DNN model output.

surpassing that of the backscatter intensity feature of 0.72, possibly due to the complex seafloor distribution of the mid-ocean ridge. As shown in Fig. 20, the reliability of acoustic backscatter intensity mapping for seafloor types is weakened by the presence of seafloor fractures and buried or semi-buried rock structures. The magnetic data with greater detection depth can effectively address the aforementioned situation and provide an important supplement for DNN models to predict accurately.

Overall, the ranking of the contributions of each feature aligns with expectations and is broadly consistent with the findings of prior related studies [62], [63]. It has been demonstrated that the classification potential of magnetic data is not inferior to that of acoustic data in the complex deep-sea environment.

C. Potential Applications and Prospects

Polymetallic nodules, as one of the representatives of deep-sea mineral resources, exhibit characteristics of being buried or semi-buried by sediment, thereby significantly increasing the difficulty of their exploration [3]. Similar to the rock structures in the complex deep-sea environments, the exposed polymetallic nodules exhibit a higher magnetic anomaly compared to the surrounding sediment. Therefore, fusing AUV acoustic and magnetic data for high-precision classification and identification of polymetallic nodules seems to be promising.

With the advancement of sensor technology, underwater hyperspectral imagers (UHIs) can now be integrated into AUVs and ROVs for conducting deep-sea surveys [64]. Spectral data with multiple bands significantly enhance the ability to characterize the seafloor, showing substantial potential in resource surveys and environmental monitoring [3]. Thus, the technical framework provided in this study also serves as a reference for integrating additional near-bottom data, such as multispectral data here.

In addition to underwater acoustic remote sensing, several advanced technologies in the field of underwater photogrammetry warrant attention, such as underwater image captioning [65], large foundation model-empowered underwater image enhancement [66], etc.; all of which demonstrate significant potential in underwater environment sensing. Image captioning technology automatically converts visual content from images into natural and fluent textual information,

effectively conveying the semantic information expressed in the image [67]. It can be expected that cross-modal learning and multimodal data fusion will also be important future directions for seafloor classification research.

VI. CONCLUSION

In response to the practical engineering exploration needs for high-precision seafloor classification in complex deep-sea environments, this study introduces a seafloor classification method that fuses AUV acoustic and magnetic data. The findings of the study are summarized as follows.

- 1) To address the issue of multisource data fusion, a spatial scale normalization model was developed. In comparison with the traditional interpolation algorithms (RMSE of 40.6 for NNI, RMSE of 27.36 for MCI, and RMSE of 25.16 for kriging interpolation), the normalized magnetic anomaly grid has the best prediction accuracy, achieving an RMSE of only 16.75.
- 2) Considering the characteristics of acoustic and magnetic heterogeneous sensors, feature-level data fusion is employed, and a DNN model based on deep learning is used for training and classification. Compared to using solely acoustic data, the integration of acoustic and magnetic data resulted in an improvement of 6.4% in the overall classification accuracy and 0.096 in the kappa coefficient.
- 3) To evaluate the importance of the magnetic features in the seafloor classification process, SHAP was employed to visualize the contribution of each feature within the DNN model. The contribution of the magnetic anomaly feature is 0.87, higher than the backscatter intensity of 0.72, second only to the depth of 1.

In addition, the proposed method still exhibits certain limitations in terms of generalizability validation and data scalability.

- 1) Although we have delineated a study area of approximately 60 km² near the SWIR hydrothermal field, the generalizability of the method remains challenging in extreme environments such as deep-sea trenches. Future expansion of sampling areas to diverse complex deep-sea scenarios could alleviate.
- 2) Although we have developed a seafloor classification framework for fusing AUV acoustic and magnetic data, its transferability to other heterogeneous data fusion scenarios (e.g., combining seafloor hyperspectral data) requires further validation. The heterogeneous data with varying spatial resolutions and nonstationary noise models may require adaptive tuning of the method parameters.

Overall, there is sufficient evidence to indicate that the integration of acoustic and magnetic data for seafloor classification is a valid and effective technical approach. Particularly in complex deep-sea environments, acoustic and magnetic data exhibit highly complementary characteristics. Future resource exploration and environmental research on the deep-sea floor will benefit from studies analogous to the seafloor classification in complex deep-sea environments.

ACKNOWLEDGMENT

The authors would like to thank the QianLong II AUV team of R/V Xiangyanghong 10 and underwater video team of Dayangyihao for collecting data.

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